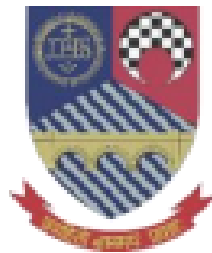


First Year (Semester-2) Research Assignment on  
*Predicting Bridge Deterioration Condition  
using Machine Learning*

in partial fulfilment of the requirement for the successful  
completion of semester 2 of MSc Big Data Analytics

Submitted By  
24-PBD-041  
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## DECLARATION

I, the undersigned solemnly declare that the research assignment *Predicting Bridge Deterioration Condition using Machine Learning* is based on my work carried out during the course of our study under the supervision of *Mr. Chetan Verma*. I assert the statements made and conclusions drawn are an outcome of my research work. I further certify that

- The work contained in the report is original and has been done by me under the general supervision of my supervisor.
- The work has not been submitted to any other Institution for any other degree / diploma / certificate in this university or any other University of India or abroad.
- We have followed the guidelines provided by the department in writing the report.

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## 1. Abstract

This research focuses on predicting the deterioration condition of bridge decks using machine learning techniques. The study utilizes key bridge attributes such as age, average daily traffic (ADT), material type, weather conditions, and width for model development. Both K-Nearest Neighbors (KNN) Regression and Random Forest (RF) algorithms were applied to analyze the relationship between these factors and the bridge deck deterioration condition. The performance of both models was evaluated using metrics like  $R^2$  score, MAE, and RMSE. The Random Forest model showed better prediction accuracy compared to KNN. This study highlights the potential of machine learning models in improving bridge condition prediction and supporting proactive maintenance planning.

## 2. Introduction

Bridges play a critical role in transportation infrastructure by connecting regions and facilitating movement of people and goods. However, with increasing age, environmental factors, and load conditions, the structural health of bridges tends to deteriorate over time, posing serious safety risks.

Recently, the Ahmedabad Municipal Corporation (AMC) has initiated a major repair and maintenance project targeting over 50 bridges in the city, some of which are nearly 50 years old. According to a report by Ahmedabad Mirror, this is one of the largest infrastructure rehabilitation drives undertaken after decades of service, highlighting the urgent need for regular monitoring and condition assessment of bridges to prevent sudden failures and accidents.

Inspired by such real-world challenges, this research focuses on developing a machine learning-based predictive model that can estimate the deterioration condition of bridges using various influencing factors like bridge age, material, traffic load, environmental conditions, and structural dimensions. Predictive modeling using machine learning can assist government bodies and infrastructure agencies in proactive decision-making, prioritizing maintenance, and optimizing resource allocation — ultimately ensuring public safety and structural longevity.

This study not only addresses the current needs of infrastructure management reflected in ongoing government initiatives but also aims to contribute to smart and data-driven bridge management systems for future cities.

### 3. Review of Literature

1. Bridge Deck Deterioration Models for Indiana Bridges Using National Bridge Inventory Data  
Jiang et al.[1] developed predictive models for bridge deck deterioration using historical bridge inspection data from Indiana. He found that the bridge deck deterioration is heavily influenced by age, traffic, material type, and environmental factors. ANN models outperformed traditional methods in predicting deterioration patterns, offering better tools for proactive bridge maintenance planning.
2. Bridge Infrastructure Asset Management System: Comparative Computational Machine Learning Approach for Evaluating and Predicting Deck Deterioration Conditions  
Rayan Assaad, S.M.ASCE<sup>1</sup> ; and Islam H. El-adaway [4] used data from the National Bridge Inventory (NBI), including features like bridge age, average daily traffic (ADT), deck area, material type, weather conditions, and span length. The author applied several ML models, including Linear Regression, K-Nearest Neighbors (KNN), Random Forest Regressor, Support Vector Regressor (SVR), and XGBoost Regressor. Data preprocessing involved handling missing values, encoding categorical variables, and feature scaling to prepare the dataset for modeling. Among the models used, Random Forest and XGBoost performed best, showing higher accuracy in predicting bridge deterioration due to their ability to capture complex patterns. The study found that factors such as bridge age, traffic volume, and environmental conditions were significant in influencing deterioration. Evaluation was done using metrics like  $R^2$  score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).
3. Development and Utilization of Bridge Data of the United States for Predicting Deck Condition Rating Using Random Forest, XGBoost, and Artificial Neural Network  
Zhao et al. (2024) [5] assessed bridge deck conditions using machine learning models with remote sensing and NBI (National Bridge Inventory) data. The dataset included bridge age, average daily traffic (ADT), deck material, surface area, climate, and remote sensing features like temperature and precipitation. They applied Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and eXtreme Gradient Boosting (XGBoost) models. Among these, Random Forest performed best with an accuracy of 83.58%. The study found that bridge age, ADT, deck material, and surface area were the most significant factors influencing deck condition. Their work demonstrated that integrating remote sensing with structural data improves the prediction of bridge deterioration.

## 4. Objective, Data & Methodology

### Objective

The primary objective of this research is to develop a machine learning model capable of predicting the deterioration condition of bridges based on multiple influencing factors.

This study aims to support smart infrastructure management by enabling early detection of vulnerable bridges and assisting authorities in preventive maintenance decisions.

Specific objectives of this study include:

- To study the key factors affecting the deterioration of bridges such as age, traffic load, material used, weather conditions, and dimensions.
- To perform exploratory data analysis to understand trends and patterns related to bridge deterioration.
- To apply and compare different machine learning algorithms (K-Nearest Neighbors and Random Forest Regressor) for predicting bridge deterioration condition.
- To perform hyperparameter tuning for improving model accuracy.
- To evaluate and compare the performance of developed models based on appropriate metrics.
- To visualize model results for better interpretation and decision-making.

### Data Collection

For this research, a structured dataset related to bridge infrastructure was used. The dataset included variables such as:

- Bridge Age
- Average Daily Traffic
- Bridge Material
- Weather Condition
- Bridge Width
- Deterioration Condition (Target Variable)

A continuous variable ranging from 0 to 9, representing the overall deterioration condition of the bridge.

- 0 → Excellent Condition
- 9 → Worst Condition (Severely Deteriorated)

Since real-time government data was not openly available, a synthetic yet realistic dataset was generated based on logical assumptions and domain knowledge. The

generated data aimed to reflect practical scenarios by considering the logical dependency of deterioration condition on factors like age, traffic, material, and environmental conditions — similar to the real-world scenario of bridges in Ahmedabad undergoing repairs after 50 years.

## Methodology

This research work aims to predict the deterioration condition of bridges using Machine Learning techniques. The following systematic methodology was adopted to carry out the study:

### 1. Data Collection

The key attributes collected include:

- Bridge Location
- Age of the Bridge
- Average Daily Traffic (ADT)
- Bridge Material Type
- Total Length and Span Details
- Deck Width
- Structural Condition Ratings (Deck, Superstructure, Substructure)

### 2. Data Preprocessing

To prepare the raw dataset for model building, several preprocessing steps were performed:

- Handling Missing Values: Missing data was either removed or imputed using appropriate statistical techniques (mean/median).
- Encoding Categorical Variables: Non-numeric variables like Bridge Material and Weather Conditions were converted into numerical formats using label encoding.
- Feature Scaling: Numerical variables were normalized to bring them to a common scale using techniques like Min-Max Scaling.
- Feature Selection: Highly correlated and relevant features were selected based on correlation analysis and domain knowledge to avoid multicollinearity.

### 3. Model Development

Different Machine Learning algorithms were applied to develop predictive models for estimating the bridge deterioration condition. The models used in this study include:

- K-Nearest Neighbors (KNN) Regressor<sup>7</sup>
- Random Forest Regressor (RF)

#### 4. Model Evaluation

The performance of each model was evaluated using standard error metrics to assess their prediction capability:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination ( $R^2$  Score)

#### 5. Model Selection

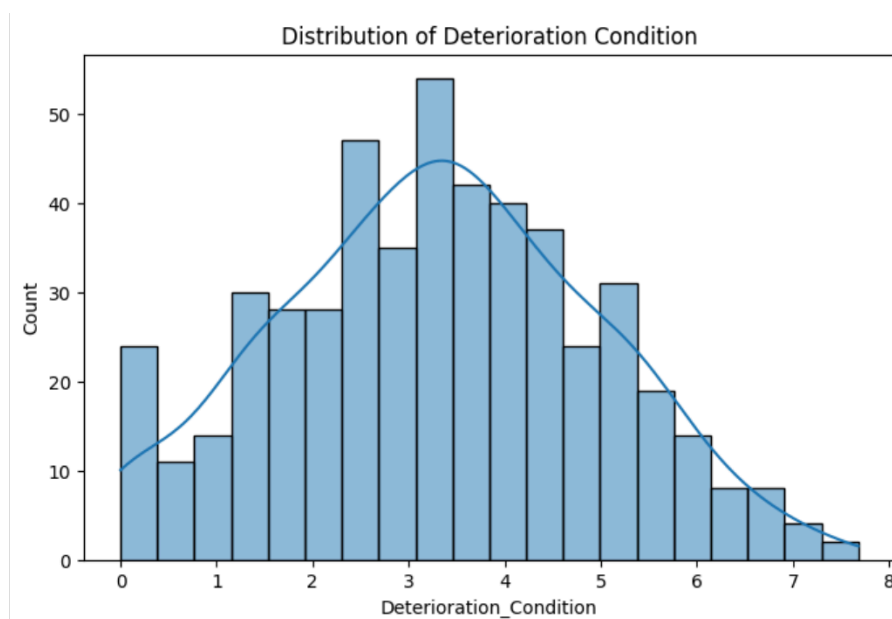
Based on the performance metrics obtained, the Random Forest Regressor showed the highest accuracy in predicting the bridge deterioration condition. It was selected as the final model for prediction due to its ability to handle non-linear data and provide high prediction accuracy.

#### 6. Prediction and Result Visualization

The final model was used to predict the deterioration condition of bridges. The comparison between actual and predicted values was visualized using various graphs for better understanding and analysis. Visualizations include:

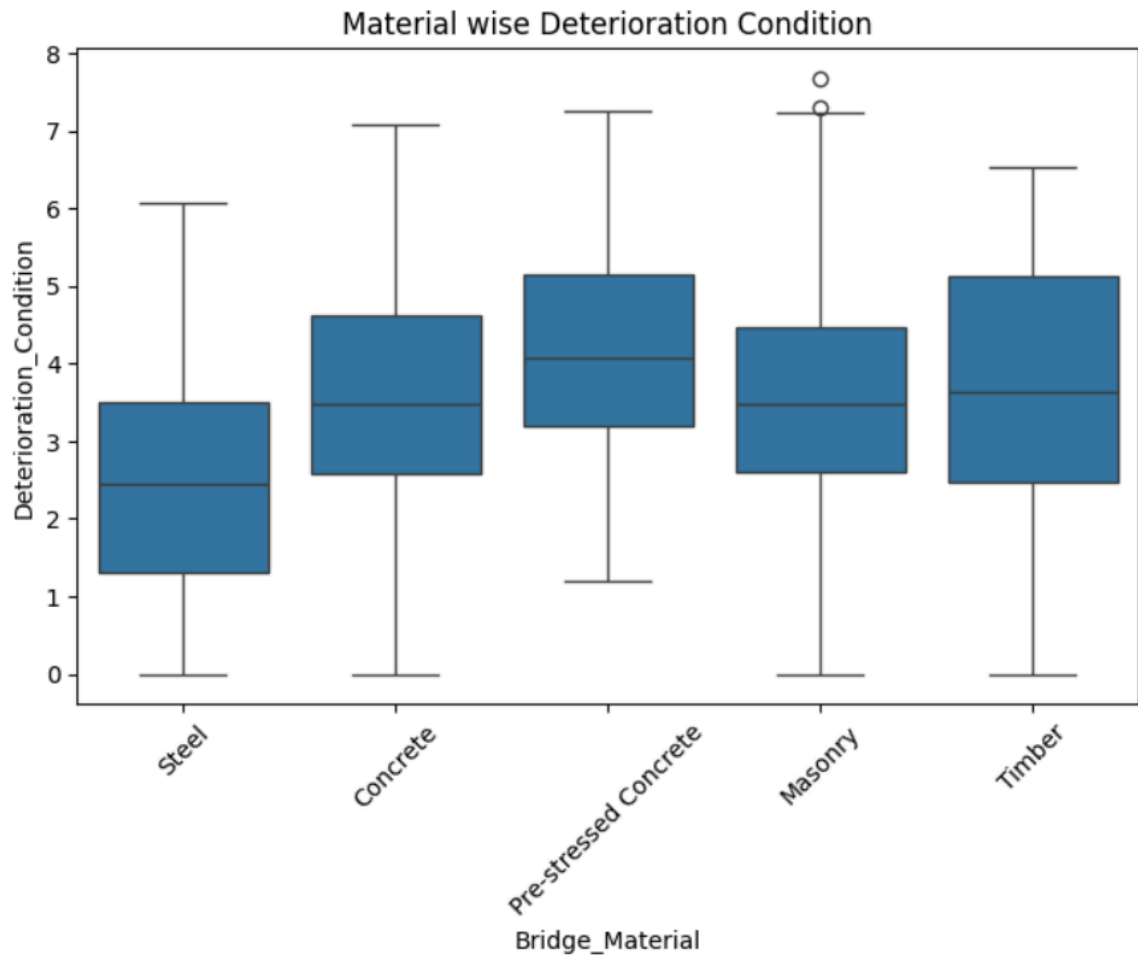
- Actual vs Predicted Plot
- Error Distribution Plot
- Feature Importance Graph

### 5. Data Analysis and Findings

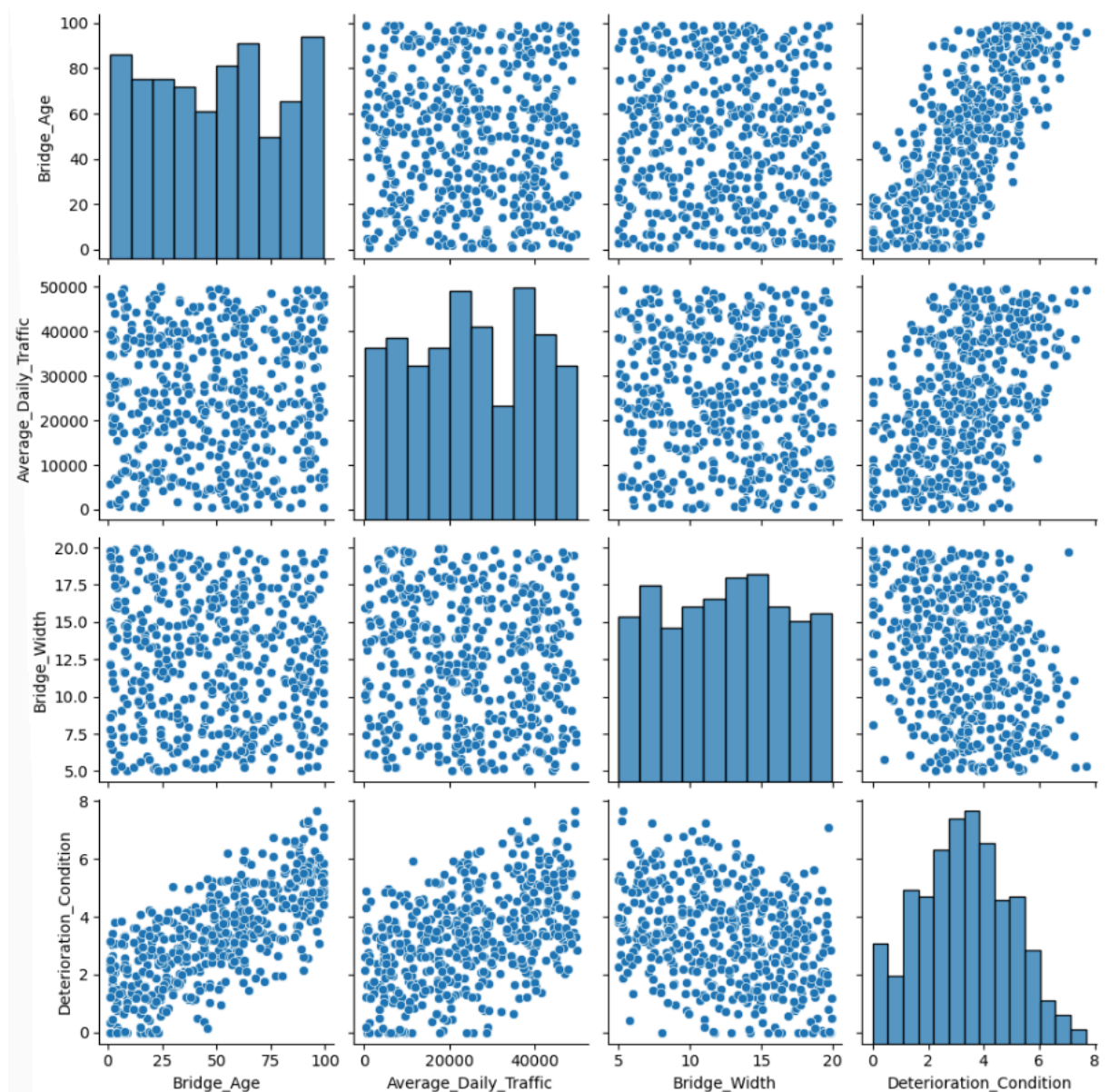




The deterioration condition of bridges predominantly ranges between 2 and 5, indicating that a significant proportion of bridges are in moderately deteriorated states. The distribution is slightly right-skewed, suggesting the presence of a few critically deteriorated bridges, but the majority fall within a manageable condition range.



Material-wise analysis revealed that Steel bridges exhibited the least deterioration, indicating higher durability and better condition retention. Conversely, Pre-stressed Concrete bridges showed the highest median deterioration, possibly due to aging or environmental factors. Masonry and Timber bridges displayed significant variability in condition, suggesting that their performance might be more sensitive to external factors.



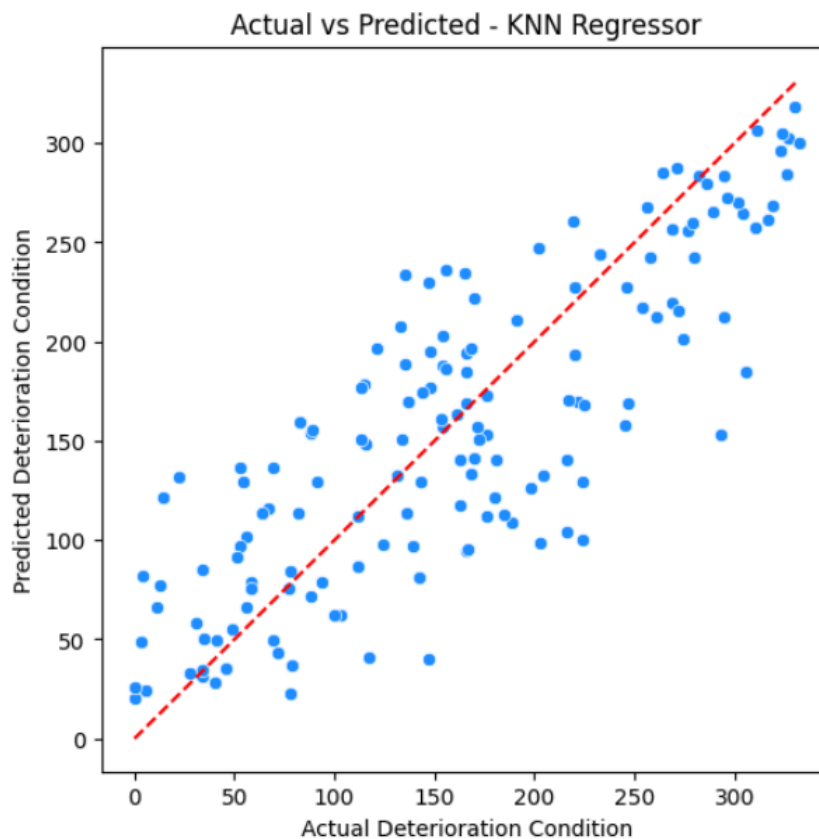
I conducted Pairplot analysis to explore the relationships between key variables: Bridge Age, Average Daily Traffic, Bridge Width, and Deterioration Condition. A positive correlation was observed between Bridge Age and Deterioration Condition, indicating that older bridges generally exhibited higher levels of deterioration. Similarly, a slight upward trend was visible between Average Daily Traffic and Deterioration Condition, suggesting that increased traffic might accelerate deterioration.

However, Bridge Width did not display any significant relationship with Deterioration Condition, implying that width alone may not directly influence structural degradation. The pairplot also highlighted considerable variability across all features, emphasizing that bridge deterioration is likely influenced by a combination of factors rather than a single variable.

## KNN Regression model

K-Nearest Neighbors Regressor was implemented to predict the deterioration condition of bridges based on the selected input variables. The model initially considered a default value of  $n\_neighbors = 5$ . The performance of the model was evaluated using  $R^2$  Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), which are essential indicators of prediction accuracy and model reliability.

The model achieved an  $R^2$  Score of 0.489, indicating that approximately 48.9% of the variance in the deterioration condition was explained by the model. The MAE was observed to be 1.32, while the RMSE was calculated as 1.69. These results reflect that although the model was able to capture certain patterns in the data, its predictive capability remained moderate. The relatively higher error values suggest that the model could benefit from further optimization of its parameters to improve accuracy and reduce prediction errors.



The Actual vs Predicted scatter plot indicates that the KNN Regression model is generally effective in predicting bridge deterioration conditions. Most predictions are reasonably accurate, particularly for moderate levels of deterioration. However, prediction accuracy tends to decline for bridges with extremely high deterioration conditions — a behavior aligned with the local averaging nature of KNN algorithms.

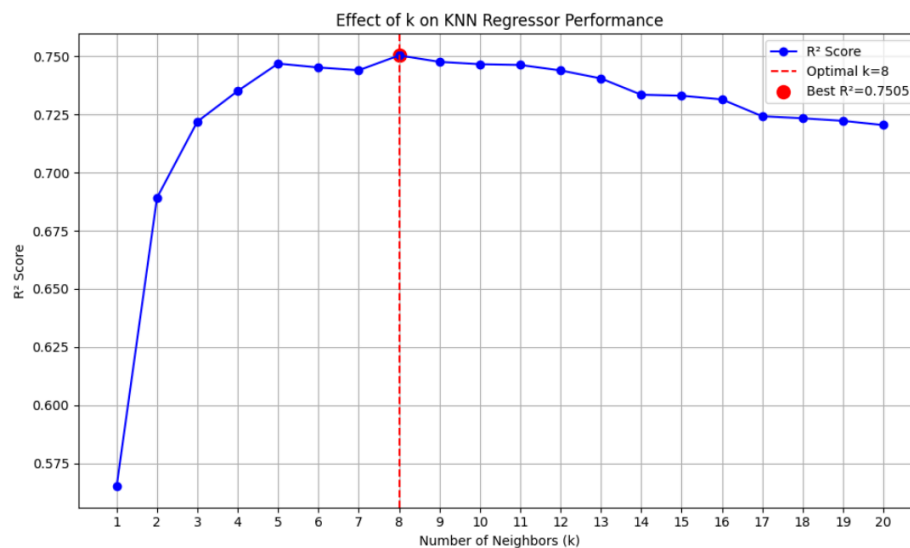
## Hyperparameter Tuning of K-Nearest Neighbors Regressor

The hyperparameter tuning of the K-Nearest Neighbors (KNN) Regression model was successfully carried out using GridSearchCV with 5-fold cross-validation. The primary objective was to identify the optimal number of neighbors ( $k$ ) that would yield the best predictive performance for estimating the bridge deterioration condition.

The tuning process revealed that the optimal value of  $k$  was 8, which led to a significant improvement in the model's  $R^2$  score from 0.6848 (with initial  $k=5$ ) to 0.7504. This indicates that the tuned model was able to explain approximately 75% of the variability present in the target variable — highlighting a substantial enhancement in model performance after hyperparameter optimization.

This improvement clearly demonstrates the importance of hyperparameter tuning in machine learning models. The optimal selection of  $k=8$  allowed the model to strike a better balance between underfitting and overfitting, thereby increasing the accuracy and robustness of the predictions.

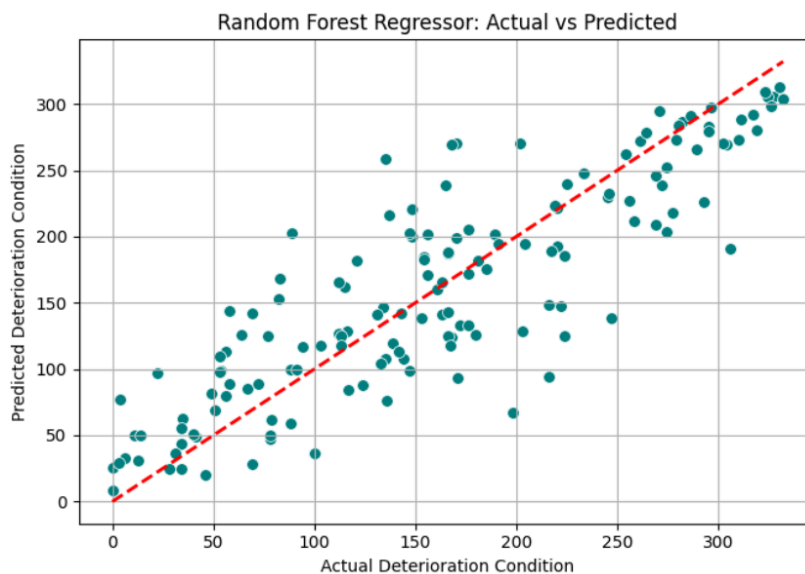
Thus, the hyperparameter-tuned KNN regression model can be considered more reliable and better suited for predicting bridge deterioration conditions compared to the initial untuned model. This tuned model forms a strong baseline for future advanced modeling or ensemble-based approaches.



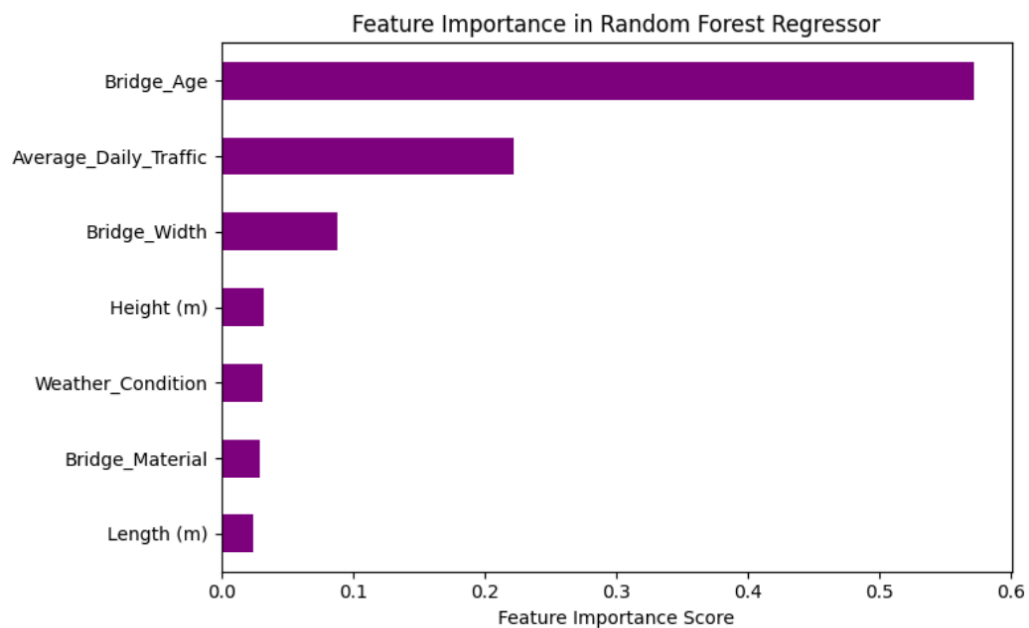
The model achieved the highest performance at  $k=8$ , after which the  $R^2$  score stabilized or slightly decreased.

## Random Forest Regressor Model

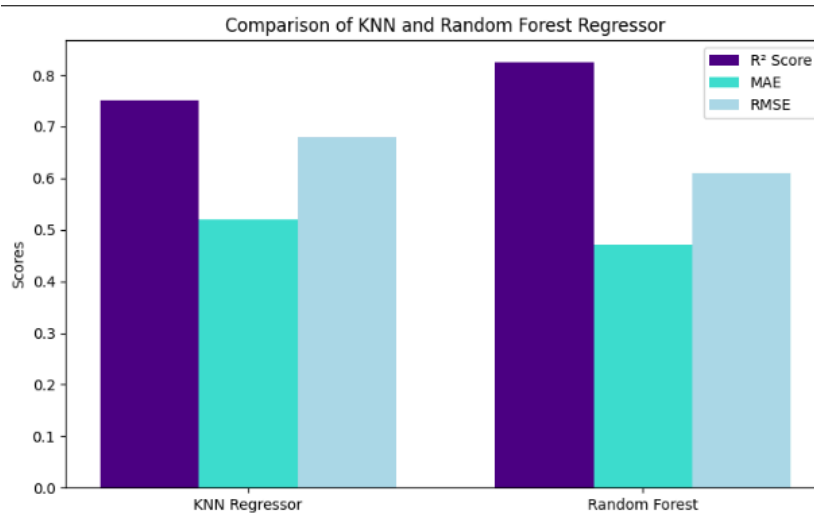
The Random Forest Regressor was implemented to predict the bridge deterioration conditions using the same dataset and features as the KNN Regressor. The model achieved an  $R^2$  score of approximately 0.8257, indicating that it can explain around 82.57% of the variability in the target variable. Additionally, the model produced a Mean Absolute Error (MAE) of 0.47 and a Root Mean Squared Error (RMSE) of 0.61, suggesting better prediction accuracy and stability.



This performance demonstrates the effectiveness of ensemble learning techniques, such as Random Forest, in handling non-linear relationships and reducing overfitting through multiple decision trees.



## Comparison between Random Forest Regressor and KNN Regressor



When comparing the performance of both models, it was observed that the Random Forest Regressor outperformed the KNN Regressor in terms of accuracy and error metrics. The KNN Regressor, after hyperparameter tuning ( $k=8$ ), achieved an  $R^2$  score of 0.7504, whereas the Random Forest Regressor achieved a higher  $R^2$  score of 0.8257.

Moreover, the error values (MAE and RMSE) of Random Forest were lower than those of the KNN model, confirming its better predictive capability on the bridge deterioration dataset. Thus, Random Forest emerges as a more robust and reliable model for predicting bridge deterioration conditions in this study.

## 6. Conclusion

This research successfully applied machine learning techniques, specifically K-Nearest Neighbors (KNN) Regression and Random Forest (RF), to predict bridge deck deterioration conditions. The dataset used in this study included essential features such as bridge age, average daily traffic (ADT), bridge material type, weather conditions, and bridge width. These factors were identified as the most influential components affecting bridge deterioration over time.

The analysis showed that older bridges with higher traffic flow, exposure to harsh weather conditions, and certain material types (like timber or steel in adverse environments) were more prone to deterioration. The Random Forest model outperformed KNN in prediction accuracy, highlighting its robustness in handling complex relationships within the data.

This study emphasizes that machine learning models can play a crucial role in monitoring bridge health, predicting deterioration trends, and supporting proactive maintenance strategies. Future research can further enhance prediction performance by incorporating larger, real-time datasets and additional structural or environmental variables.

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